TRACK 1: VARIANT PREDICTION

Mosaic Variegated Aneuploidy (MVA) Hackathon 2026 • Methods & Model Report

Team: Los Tanos
Model: Model 1 (Automated Pipeline)
Genome: GRCh38 / hg38

Abstract & Method Overview: We developed an automated computational pipeline to prioritize causal genetic variants from whole-genome sequencing (WGS) data. The framework integrates strict sequencing quality filters and background allele-aware depletion with local snpEff annotation, ontology-aware Resnik semantic similarity, and multi-parametric compound-heterozygous modeling. Programmatically evaluating 5,012,204 initial variants in ~60 seconds, our pipeline isolated a biallelic compound-heterozygous pair in **BUB1B** as the primary driver with an EPCR score of **0.9379**.

5.01M	591k	905	0.9379
RAW INPUT VARIANTS	RARE HIGH-QC (88.2% DROP)	PROTEIN ALTERING	TOP CAUSAL EPCR (BUB1B)

1. FIVE-STAGE COMPUTATIONAL ARCHITECTURE

The algorithmic architecture executes sequentially without human intervention, balancing background noise filtration with the retention of genuine pathogenic alleles:

STEP 1
Phenotype Extraction
Extracted 8 HPO terms, anchoring on Rhabdomyosarcoma and recurrent miscarriages.

STEP 2
VCF Pre-Filtering
Enforced GQ≥20, DP≥10, PASS, AB 0.30–0.70, and allele-aware dbSNP common subtraction.

STEP 3
Annotation Tiering
Built local snpEff GRCh38.99, queried MyVariant & AlphaGenome; enforced gnomAD AF < 1%.

STEP 4
Phenotype Matching
Ontology-aware Resnik semantic scoring. BUB1B emerged #2 genome-wide (score 0.632).

STEP 5
Ranking & EPCR
Dynamic compound-het pairing and unified EPCR scoring. BUB1B prioritized confidently.

2. METHODOLOGY DETAILS & FILTERING LOGIC

- **Phenotypic Anchoring (Step 1):** Clinical terms were mapped directly to ontology anchors: $HP{:}0002859$ (Rhabdomyosarcoma) and $HP{:}0200067$ (Recurrent spontaneous abortion), focusing downstream semantic weighting toward aneuploidy and mitotic checkpoint loci.
- **Quality Control & Background Depletion (Step 2):** Raw calls were filtered by intrinsic metrics ($GQ \geq 20$, $DP \geq 10$, $0.30 \leq AB \leq 0.70$ for heterozygous positions). Allele-aware subtraction against dbSNP common (b151, $MAF \geq 0.01$) eliminated 88.2% of raw calls, reducing 5,012,204 variants to 591,339 high-confidence rare variants.
- **Deleteriousness & Frequency Pruning (Step 3):** Using local Ensembl 99 GTF/FASTA resources and programmatic APIs, 905 protein-altering (HIGH/MODERATE impact) variants were isolated. Any variant with gnomAD exome/genome $AF > 1\%$ was excluded.
- **Ontology-Aware Prioritization (Step 4):** Candidate genes with qualified variants were evaluated against the patient's HPO profile using Resnik semantic similarity. *BUB1B* scored 0.632, representing the sole Tier 1 MVA panel gene with protein-altering rare variants.

Submission Nature & Automation

The submission file represents the **100% automated output** of our pipeline. Ranking, pairing, and EPCR calculations were computed programmatically. Downstream clinical curation was strictly observational to validate biological plausibility.

3. PRIMARY VARIANT FINDINGS & BIOLOGICAL CONCORDANCE

The pipeline identified a compound-heterozygous pair in **BUB1B** as the primary causal finding, establishing complete biallelic disruption. The candidate achieved an EPCR of **0.9379**, exhibiting decisive separation from the monoallelic runner-up (*LZTR1*, EPCR = 0.5627).

Attribute	Allele 1 (Primary Variant 1)	Allele 2 (Primary Variant 2)
Gene & OMIM	*BUB1B* (OMIM *602860 / MVA Syndrome 1 #257300)	*BUB1B* (OMIM *602860 / MVA Syndrome 1 #257300)
Coordinates (GRCh38)	chr15:40209701 (ref_1: T, alt_1: A)	chr15:40220612 (ref_2: T, alt_2: C)
Molecular Consequence	Exon 17: Stop-gain (TTA → TGA, p.Leu751Ter)	Exon 23: Missense (AAT → AAG, p.AsnLys)
Impact Tier	HIGH (Premature truncation / Nonsense-mediated decay)	MODERATE (Catalytic/Checkpoint domain disruption)
Pathogenicity Scoring	CADD: 36.0 | ClinVar: Pathogenic / Likely Pathogenic	REVEL: 0.812 | AlphaMissense: Likely Pathogenic
Finding Classification	**Compound-Heterozygous Biallelic Pair — Joint EPCR: 0.9379 (Primary Finding)**	

Biological Rationale & Phenotype Correlation

BUB1B encodes the spindle assembly checkpoint (SAC) kinase BUBR1, which inhibits the APC/C complex to prevent premature chromosome segregation. Biallelic loss of functional BUBR1 causes **Mosaic Variegated Aneuploidy Syndrome 1**, characterized by high-rate aneuploidies, severe growth deficiency, microcephaly, and elevated childhood embryonal tumor predisposition (specifically **rhabdomyosarcoma** and Wilms tumor). Parental subfertility and recurrent pregnancy losses are established manifestations, matching the extracted HPO terms (***HP:0002859*** and ***HP:0200067***).

4. DOWNSTREAM CURATION, VERIFICATION & ETHICAL DESIGN

- **Quality Auditing:** A tabix check of 300 dbSNP positions revealed coverage gaps in legacy common sets, validating our downstream gnomAD $AF < 0.01$ cutoff.
- **Structural Impact:** Truncation at residue 751 deletes the C-terminal kinase domain, while the exon 23 missense variant impairs residual spindle checkpoint fidelity.
- **Ethical Boundaries & Findings:** Non-coding regions and incidental findings were not reported to minimize false discovery. Variants in genes lacking phenotype convergence (e.g., *SGO1*, *MAD1L1*) were penalized by the semantic scoring model.

5. GENERATIVE AI SPECIFICATION & DATA GOVERNANCE

Compliance Notice (Update 28 Aug 2026)

Generative AI Providers & Settings: Google API (Gemini Pro) and Anthropic API (Claude Opus) were accessed under *commercial tier enterprise terms* with *strict zero-retention data-handling settings and no model training on customer content*.

Public Data: Ensembl Rel 99, dbSNP b151, gnomAD v3/v4, HPO (2026-09-01), MyVariant.info, AlphaGenome Atlas, ClinVar, OMIM. **Proprietary Data:** None.

6. PIPELINE EVALUATION, STRENGTHS & LIMITATIONS

Key Strengths

- Fully automated, deterministic, and reproducible.
- Native compound-heterozygous allele pairing.
- Multidimensional integration of in-silico & HPO metrics.

Known Limitations

- Single-sample VCF cannot confirm *in trans* phase.
- Dependent on curated HPO ontology coverage.
- Non-coding regulatory variants currently excluded.

Compute Performance: Pipeline run time ≈ 60 seconds (with pre-cached databases) on 2 CPU cores, 8 GB RAM, and 8 GB disk space. Total infrastructure cost: $0.00.